

Welfare, Integrability and Afriat

Advanced Microeconomics 1: Economic Decision Making. Lecture 4

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The state of play

The question: Who did inflation hurt, is wage indexation the right policy response?

1. **L1**: we can formally pose the question.
2. **L2**: the ideal answer assuming u is $e(p^1, u^0) - w$, per household.
3. **L3**: the building blocks for not assuming u .
4. **Today**: (1) putting together the pieces $\rightarrow$ finding the answer from data, (2) interrogate the remaining assumptions.

Recovering CV using demand functions

The integrability problem

Given data a demand function $x(p, w)$ what can we learn about the underlying preference relation?

Theorem · integrability (MWG 3.H)

If $x(p, w)$ satisfies Walras' law, h.d.0, and has a symmetric, negative semidefinite Slutsky matrix, then there exists a rational preference relation rationalising x .

Proof → p. 27

Proof: sketch in the appendix; full treatment MWG 3.H.

An alternative approach: Revealed Preferences

Definition · SARP (Houthakker 1950)

$x(p, w)$ satisfies **SARP** if for any chain $(p^1, w^1), \dots, (p^N, w^N)$ with $x(p^{n+1}, w^{n+1}) \neq x(p^n, w^n)$ and $p^n \cdot x(p^{n+1}, w^{n+1}) \leq w^n$ for all $n \leq N - 1$, we have $p^N \cdot x(p^1, w^1) > w^N$.

Theorem · Houthakker (1950)

$x(p, w)$ satisfies SARP $\iff$ there is a rational preference relation that rationalizes x .

Proof deferred (Houthakker 1950; MWG 3.J).

One difference: SARP doesn't need the derivatives of x to exist.

What does this theorem get us?

1. Find a demand function x such that:

1.1 The following hold (1) h.d.0, (2) Walras, (3) symmetric NSD S

1.2 OR $x(p, w)$ satisfies SARP

⇒ there exists a rational u that generated x .

⇒ x is the product of optimisation.

⇒ v, e, h exist and we can use Shephard, Slutsky, and Roy.

⇒ we can then recover v and thus e , and thus CV .

In practice

1. Check your demand function $x(p, w)$ is h.d.0, satisfies Walras and its S is symmetric NSD.
2. Define $w(p) = e(p, u^0)$ the wealth needed to maintain u^0 as some p changes.
3. **Shephard** gets you $h(p, u^0) = \frac{de(p, u^0)}{dp}$.
4. Duality implies $h(p, u^0) = x(p, e(p, u^0)) = x(p, w(p))$.
5. Thus we have constructed the ODE: $\frac{dw(p)}{dp} = x(p, w(p))$.
6. Solve: $-CV = e(p^1, u^0) - w = \int_{p_0}^{p^1} x(p, w(p)) dp$

But we need a demand function

- So we have our answer, CV , estimable without assuming a utility function
$$-CV = \int_{p_0}^{p_1} x(p, w(p)) dp.$$
- But, we still need a demand function $x(p, w)$.
- We don't have to assume this, we can fit it to data.
- Enter econometrics!

Taking this to our question

In practice, for Belgium 2022: the ingredients

- **The shock:** as before home energy +57.9% over 2022 (NBB), so $g = 0.579$.
- **Demand function:** *Stone–Geary*, $x_E(p, w) = \gamma + \frac{\beta}{p}(w - p\gamma)$.
- **The elasticity:** residential energy demand is *price-inelastic*. Belgian causal estimate (electricity, VAT-reform quasi-experiment): $\eta \in [-0.17, -0.09]$ (Hindriks and Serse 2022). We calibrate (β, γ) so the form's share and elasticity match (s_0, η) at p^0 .

In practice, for Belgium 2022: the answer

Solving $\frac{dw}{dp} = \gamma(1 - \beta) + \frac{\beta}{p} w$ (linear; integrating factor $p^{-\beta}$):

$$\frac{-CV}{w} = \frac{\gamma p^1 + (w - \gamma p^0) (p^1 / p^0)^\beta}{w} - 1$$

With $g = 57.9\%$, and $\eta = -0.13$, recovered $-CV/w$:

	fixed basket (Laspeyres)	data elasticity (Stone–Geary, $\eta = -0.13$)	Cobb–Douglas ($\gamma = 0$)
poor, $s_0 = 0.10$	5.8%	5.7%	4.7%
rich, $s_0 = 0.06$	3.5%	3.4%	2.8%

Checking our assumptions

With a demand function $x(p, w)$ in hand

x is generated by some rational preference relation iff...

1. x is (i) h.d.0, (ii) Walras, (iii) symmetric NSD Slutsky matrix.
2. x satisfies SARP.

Non-parametric test with just data $\{(p^t, x^t)\}_{t=1}^T$, no estimated demand function

Let R denote “revealed preferred” i.e. $x^t R x^s$ means: I chose x^t even though x^s was (weakly) available i.e. $p^t \cdot x^s \leq w^t$. P denotes the strict version.

Definition · GARP

No chain $x^{t_1} R x^{t_2} R \dots R x^{t_n}$ with $x^{t_n} P x^{t_1}$ exists.

Theorem · Afriat (1967)

Data satisfy GARP $\Leftrightarrow$ Data is rationalisable by a continuous, increasing, concave utility function.

Proof $\rightarrow$ p. 28

Checking in practice

- For our worked question we can check the rationality of our demand function.
- In the practical session you will check your own rationality using GARP and Afriat.

Rationality of our demand function

Energy demand $x_E = \gamma(1 - \beta) + \frac{\beta w}{p_E}$; other good $x_O = \frac{(1-\beta)}{p_O}(w - p_E\gamma)$. Write $y = w - p_E\gamma$.

1. **h.d.0.** Scale $(p_E, p_O, w) \rightarrow (tp_E, tp_O, tw)$, run-through, t 's cancel ✓

2. **Walras' law.** $p_E x_E + p_O x_O = (p_E\gamma + \beta y) + (1 - \beta)y = p_E\gamma + y = w$. ✓

3. **Slutsky symmetric & NSD.** $S_{\ell k} = \frac{\partial x_\ell}{\partial p_k} + \frac{\partial x_\ell}{\partial w} x_k$. Compute the cross terms:

$$S_{EO} = \frac{\beta(1 - \beta) y}{p_E p_O} = S_{OE}, \quad S_{EE} = -\frac{\beta(1 - \beta) y}{p_E^2} < 0, \quad S_{OO} = -\frac{\beta(1 - \beta) y}{p_O^2} < 0.$$

Own-price terms negative, and $\det S = 0 \Rightarrow S$ is negative semidefinite. ✓

Our assumptions

1. **Rationality** — choices reveal a stable ranking. *Tested:* GARP, CCEI. On you.
2. **No money illusion** — h.d.0, filed away on the first content slide of Lecture 1.
3. **Stable preferences** — the u^0 being restored is the same object across the comparison.
4. **A choice of reference** — CV or EV; we treated it as bookkeeping.

Assumption 1: rationality — the evidence

Does the observable footprint (GARP) actually hold when we look?

- **For:** in the largest test — ~2,000 Dutch adults, portfolio choices — consistency far exceeds random-choice benchmarks (Choi, Kariv, Müller & Silverman, AER 2014). Rationality is not universal, but it is the modal behaviour..
- **Against:** consistency varies sharply across people — lower for the poor, less educated, older, and those under cognitive load or stress (Kim et al., *Science* 2018; CKMS 2014). Never holds exactly and always.
- In the practical we will test it on this everyone here...

Assumption 2: no money illusion — the evidence

h.d.0 says scaling all prices and wealth changes nothing.

- **For:** over long horizons and with salient information, real variables always dominate.
- **Against:** in the short run, no. Nominal framing systematically moves choices — Shafir, Diamond & Tversky (QJE 1997) show people reason in nominal terms about wages, contracts and prices even when told the inflation rate. A 2% wage raise at 4% inflation *feels* like a gain.
- **Why it matters here:** indexing exists because people worry about the “real”.

Assumption 3: stable preferences — the evidence

The u^0 we restore must be the *same* object before and after the shock. Is it?

- **Against:** preferences are often *constructed at the moment of choice*, not retrieved from a fixed ranking. Normatively-equivalent elicitations give systematically different answers — **preference reversals**: people choose A over B yet price B above A (Lichtenstein & Slovic 1971, replicated on a Las Vegas casino floor; survives incentives, Grether & Plott 1979). If the ranking depends on how you ask, “the same u^0 ” is not innocent.
- **For (the defence):** reversals are largest in unfamiliar, one-shot lab tasks. For *repeated, familiar* decisions — groceries, energy, the household budget — choices are far more stable, which is exactly the domain our recovery lives in.
- **The honest position:** stability is a reasonable *working* assumption for routine consumption, and a fragile one for novel or framed choices. The 2022 shock is novel — a caveat worth carrying.

Assumption 4: the reference point — the evidence

CV and EV have different reference points, this shouldn't matter...

- **Against:** people value losses from a reference far more than equivalent gains — **loss aversion** (Kahneman & Tversky 1979). So the *endowment effect*: willingness-to-accept to give up a good routinely exceeds willingness-to-pay to acquire it, by a factor of ~ 2 (Kahneman, Knetsch & Thaler, JEP 1991).
- **Why it matters here:** which reference the policy adopts — the old standard of living (u^0 , CV) or the new (u^1 , EV) — now matters.

Reference points

- “You own the mug — minimum price to sell?” [insert: class median WTA]
- “You don’t — maximum price to buy?” [insert: class median WTP]
- The reference point should not matter your answers should coincide... [insert: class ratio]. From the lit: $WTA \approx 2 \times WTP$.

Behavioural Economics

Kahneman, Knetsch & Thaler (JEP 1991): “The Endowment Effect, Loss Aversion, and Status Quo Bias”, enter **behavioural economics**.

Where does that leave us

1. Rationality. **OK on average** — but heterogenous along income, education, stress, etc.
2. No money illusion. **Holds in the long-run.**
3. Stable preferences. **Holds more for regular than rare purchases**, verdict still fairly open.
4. Reference point. **Dead.** Certainly matters, how much is a little unclear.

We can extend the framework to allow for deviations to each assumption — for another day :)

Summing up

The block's scorecard

Takeaways

- **(a) Who did the 2022 energy shock hurt, and by how much?** Everyone by 3-6%, the relatively poorer by more.
- **(b) Is one-basket indexation the right compensation?** No — it over-pays the average and misses both tails.
- **Next:** Firms, where do those price rises come from?

Appendix: proofs

Proof sketch: integrability (MWG 3.H)

Step 1 — recover e from x . Fix (p^0, w^0) and try to solve the system of PDEs given by Shephard run backwards:

$$\nabla_p e(p) = x(p, e(p)), \quad e(p^0) = w^0.$$

A solution exists locally iff the system is consistent — Frobenius' condition: the “cross-derivatives” of the right-hand side must commute. Computing them, the condition is exactly **symmetry of $S(p, e(p))$** . That is the moment symmetry stops being decoration: it is the solvability condition for reconstructing the cost of living from demand.

Step 2 — recover preferences from e . Given $e(\cdot, u)$ (concave, by NSD, in the recovered system), define upper contour sets as the intersections of half-spaces

$$V_u = \{x \in \mathbb{R}_+^L : p \cdot x \geq e(p, u) \text{ for all } p \gg 0\}.$$

Concave-duality (support-function) arguments show the preference relation these sets induce is rational and generates x . □ (sketch; full argument MWG 3.H.1–3.H.2)

Proof sketch: Afriat's theorem

(iii) $\Rightarrow$ (i): a rationalizing utility orders the data; revealed-preference chains inherit the order, so no cycle can end strictly above its start.

(i) $\Rightarrow$ (ii): GARP makes the revealed relation acyclic, so observations can be ordered; a combinatorial induction then picks $U_t, \lambda_t > 0$ satisfying the inequalities — read as “a concave function with slope $\lambda_t p^t$ at x^t leaves room for U_s at x^s .”

(ii) $\Rightarrow$ (iii): build the utility explicitly as the lower envelope of those supporting hyperplanes:

$$u(x) = \min_t \{ U_t + \lambda_t p^t \cdot (x - x^t) \}$$

— continuous, increasing, concave by construction, and it rationalizes every observation.



Sketch; Afriat (1967), Varian (1982). The CCEI relaxes (i): your Thursday number is the largest budget scaling $e^* \leq 1$ at which GARP holds.

Derivation: solving the Stone–Geary recovery ODE

Stone–Geary energy demand $x_E(p, w) = \gamma + \frac{\beta}{p}(w - p\gamma) = \gamma(1 - \beta) + \frac{\beta}{p} w$, so the recovery ODE $\frac{dw}{dp} = x_E(p, w(p))$ is

$$\frac{dw}{dp} - \frac{\beta}{p} w = \gamma(1 - \beta).$$

Linear, first order — integrating factor $\mu(p) = \exp\left(-\int \frac{\beta}{p} dp\right) = p^{-\beta}$. Multiply through:

$$\frac{d}{dp}\left(w p^{-\beta}\right) = \gamma(1 - \beta) p^{-\beta} \implies w p^{-\beta} = \gamma p^{1-\beta} + C.$$

So $w(p) = \gamma p + C p^{\beta}$. **Pin C from the start:** $w(p^0) = w$ gives $C = (w - \gamma p^0)(p^0)^{-\beta}$, hence

$$w(p) = \gamma p + (w - \gamma p^0) \left(\frac{p}{p^0}\right)^{\beta}.$$

Evaluate at p^1 : $-CV = w(p^1) - w$, and $\frac{-CV}{w} = \frac{\gamma p^1 + (w - \gamma p^0)(p^1/p^0)^{\beta}}{w} - 1$.

Derivation: calibrating (β, γ) to the data

Normalise $p^0 = 1, w = 1$). Targets: the energy share s_0 and the own-price elasticity η . Stone–Geary energy demand $x_E = \gamma(1 - \beta) + \frac{\beta w}{p}$, so at the initial point $x_E^0 = \gamma + \beta(1 - \gamma)$.

Target 1 — share. $s_0 = \frac{p x_E}{w} = x_E^0 = \gamma + \beta(1 - \gamma)$.

Target 2 — elasticity. $\frac{\partial x_E}{\partial p} = -\frac{\beta w}{p^2}$, so $\eta = \frac{\partial x_E}{\partial p} \cdot \frac{p}{x_E} = -\frac{\beta}{x_E^0} = -\frac{\beta}{s_0}$.

Target 2 gives β directly; substitute into Target 1 for γ :

$$\beta = -\eta s_0, \quad \gamma = \frac{s_0(1 + \eta)}{1 + \eta s_0}.$$

- Poor ($s_0 = 0.10, \eta = -0.13$): $\beta = 0.013, \gamma = 0.088$. So $\gamma/s_0 \approx 88\%$ of energy is *subsistence* — the can't-cut-it floor. Only $\sim 12\%$ is price-responsive: that *is* the inelasticity.